

Module: Problem 4

This module provides a single utility function that determines whether a given positive integer (or an integer-compatible value) is a triangular number. A triangular number T_n satisfies the formula $T_n = n(n+1)/2$. The implementation uses the inverse quadratic formula to test the property.

Functions

`is_triangular(k)`

Description:

Evaluates whether the argument `k` corresponds to a triangular number. The function computes

$$[x = \frac{\sqrt{8k+1} - 1}{2}]$$

and returns `True` if `x` is an integer (i.e., `k` can be expressed as $n(n+1)/2$ for some integer n), otherwise `False`.

Parameters: - `k` (*int or bool*): The value to test. Expected to be a non-negative integer; booleans are accepted because they are subclasses of `int` in Python.

Returns:

`bool` - `True` if `k` is triangular, `False` otherwise.

Examples:

```
# Observed runtime behavior
result = is_triangular(0)      # → True  (note: 0 is not a standard triangular number)
result = is_triangular(1)      # → True
result = is_triangular(True)   # → True
```

Edge Cases / Notes: - **Negative inputs:** `is_triangular(-1)` raises `ValueError: math domain error` because the expression under the square root becomes negative. - **Non-numeric inputs:** Passing a string such as `'test'` or `''` triggers `TypeError: can only concatenate str (not "int") to str`. This arises from the attempt to perform arithmetic with a non-numeric type. - **Zero handling:** The function returns `True` for `k = 0`, even though 0 is not traditionally considered a triangular number. Users may need to add an explicit check if a strict definition is required. - **Missing import:** The function relies on `math.sqrt`; ensure `import math` is present in the module before calling the function.